

Quiz #3

1. Suppose X is a continuous random variable.

(a) What is $P(a \leq X \leq a)$?

$$P(a \leq X \leq a) = 0.$$

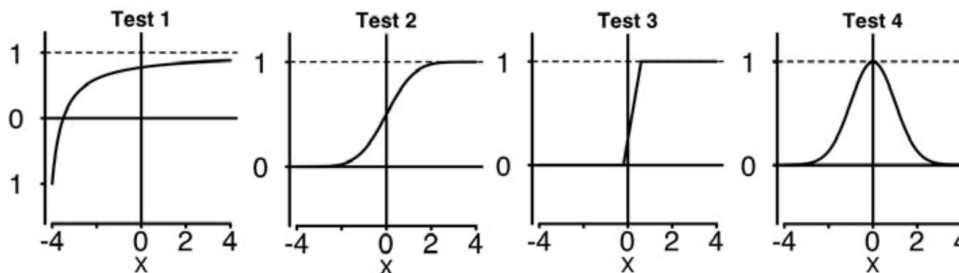
(b) What is $P(X = 0)$?

$$P(X = 0) = 0.$$

(c) Does $P(X = 2) = 0$ mean that X never equals 2?

No. For a continuous random variable, the probability of any single exact value is zero. Only an interval (range) of values can have nonzero probability.

2. Which of the following are valid cumulative distribution functions? Why are the others not valid?



Test 2 and Test 3. Test 1 is not a cdf: it takes negative values, but probabilities are positive. Test 2 is a cdf: it increases from 0 to 1. Test 3 is a cdf: it increases from 0 to 1. Test 4 is not a cdf: it decreases. A cdf must be non-decreasing since it represents accumulated probability.

3. Brrrr. Suppose the mean temperature in Pittsburgh for January was 41°F and the variance was 16°F^2 . The conversion formula between Celsius and Fahrenheit is

$$C = \frac{5}{9}(F - 32).$$

- (a) What was the standard deviation of the temperature in °F?

$$\sigma_F = \sqrt{\text{Var}(F)} = \sqrt{16} = 4^\circ\text{F}.$$

- (b) What was the mean temperature in °C?

$$\mu_C = \frac{5}{9}(41 - 32) = \frac{5}{9}(9) = 5^\circ\text{C}.$$

- (c) What was the standard deviation of the temperature in °C?

Since

$$C = \frac{5}{9}F - \frac{160}{9},$$

subtracting a constant does not change the standard deviation, while multiplying by $\frac{5}{9}$ multiplies the standard deviation by $\frac{5}{9}$. Therefore,

$$\sigma_C = \frac{5}{9}\sigma_F = \frac{5}{9}(4) = \frac{20}{9} \approx 2.22^\circ\text{C}.$$

4. Suppose exam scores X are approximately normal with mean 70 and standard deviation 4. In words, explain what each line of code calculates. You do not need to compute the numerical values.

```
from scipy.stats import norm
```

```
p1 = norm.cdf(74, loc=70, scale=4)
```

```
p2 = 1 - norm.cdf(74, loc=70, scale=4)
```

```
p3 = norm.cdf(74, loc=70, scale=4) - norm.cdf(66, loc=70, scale=4)
```

Here, `loc=70` specifies the mean and `scale=4` specifies the standard deviation.

- `p1` is the probability that a score is at or below 74: $P(X \leq 74)$.
- `p2` is the probability that a score is above 74: $P(X > 74)$.
- `p3` is the probability that a score is between 66 and 74: $P(66 < X \leq 74)$.

Because X is continuous, including or excluding the endpoints does not change these probabilities.